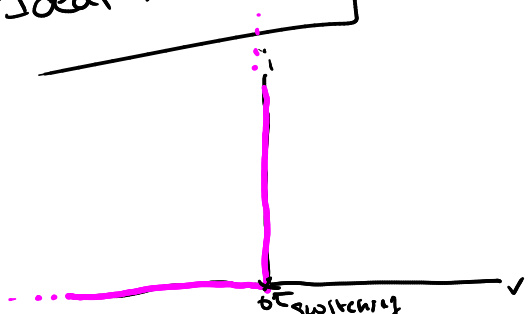
Lecture 11?

# Diodes

When $i > 0$, then $v = 0$

* Forward Bias
* Conducting
* ON

## *Ideal Diode Model



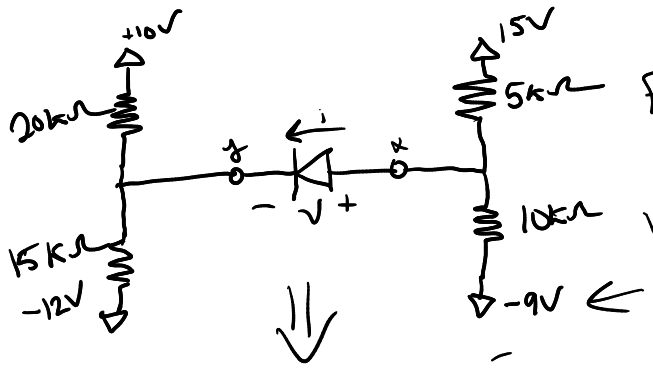

When $v < 0$
$i = 0$

* Reverse Bias
* cutoff
* non conducting
* OFF

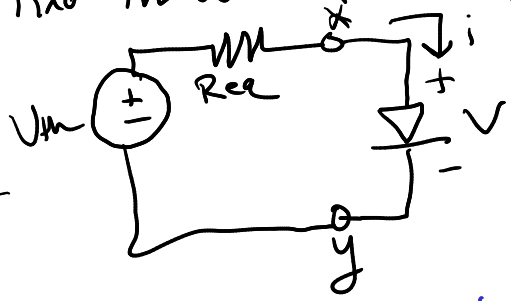
Given: Ideal Diode Model

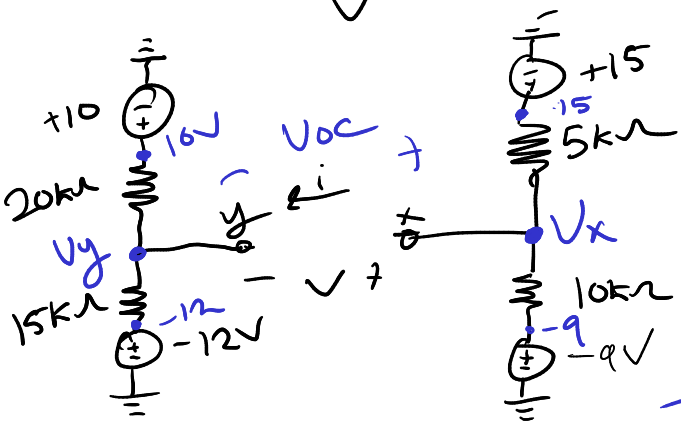
Find $v$ and $i$

Example)




Find Thévenin Eq. Seen by diode




$V_{th} = V_{oc}$ $\quad V_{oc} = V_x - V_y$

@Nodal $V_x$

$$\frac{1}{5000}(V_x - 15) + \frac{1}{10{,}000}(V_x - (-9)) = 0$$

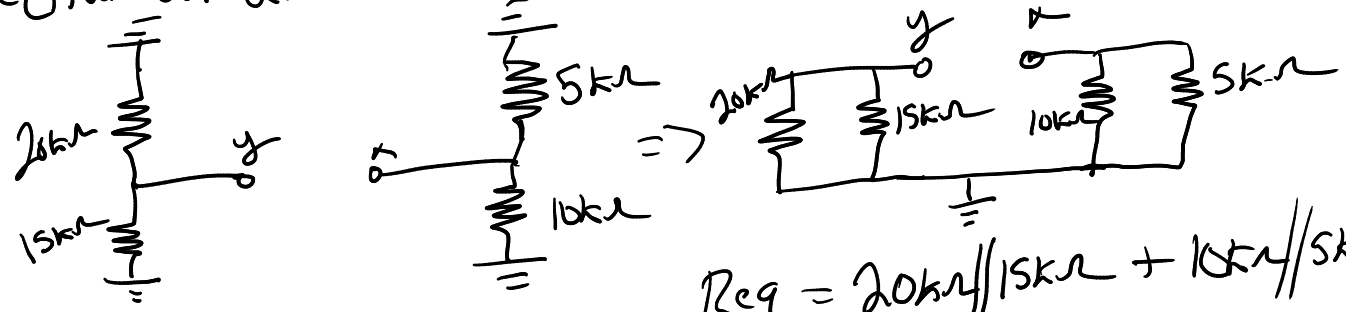
@$V_y$

$$\frac{1}{20{,}000}(V_y - 10) + \frac{1}{15{,}000}(V_y - (-12)) = 0$$

$V_x \approx 7V$ $\quad V_y \approx -2.570V$

$V_{oc} = V_x - V_y = 9.57V$

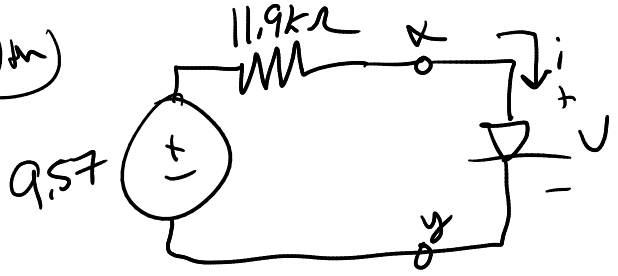
For $R_{eq}$, turn off all sources



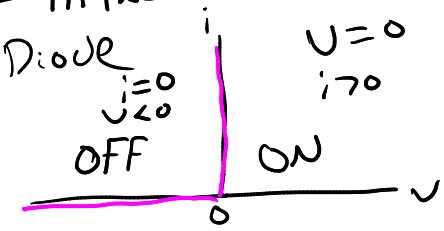

$R_{eq} = 20k\Omega \| 15k\Omega + 10k\Omega \| 5k\Omega$

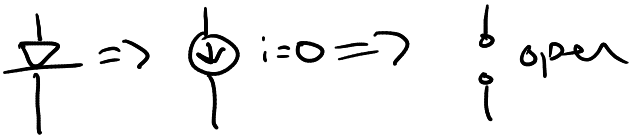
$R_{eq} = 11.9k\Omega$

$V_{th}$)

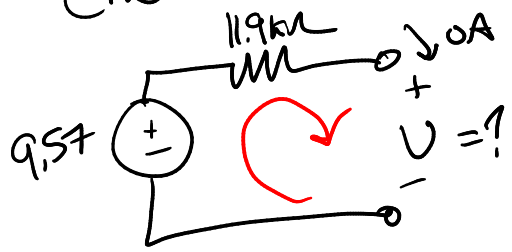



Ideal Diode



Assume Diode is off ... $i = 0$  open

Check $V < 0$?




$-9.57 + 0 + V = 0$

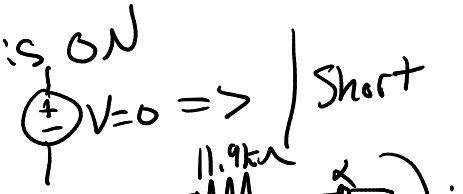
$V = 9.57$

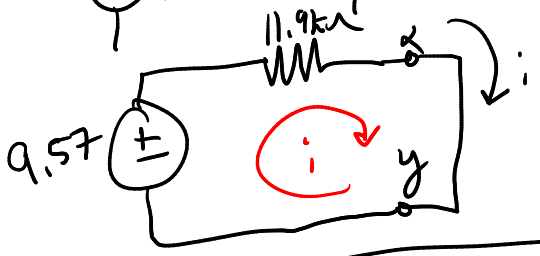
Is $V < 0$? NO

Diode is NOT OFF

Check if $i > 0$?

Assume Diode is ON

$V = 0 =$  $V = 0 \Rightarrow$ Short



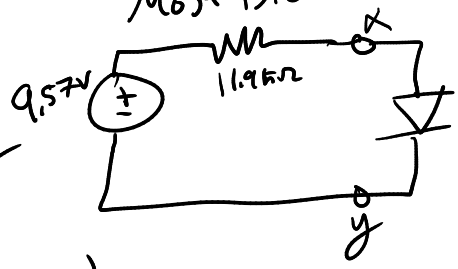

$-9.57 + 11.9 \times 10^3 \, i = 0$

$i = 804 \mu A$

$$\boxed{V = 0V, \; i = 804 \mu A}$$

*Constant Voltage Drop Model (CVD)

Most Diodes $\cong 0.7V$



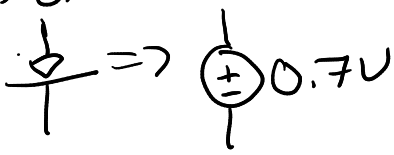

If diode is off,

Replace $\Rightarrow$ open

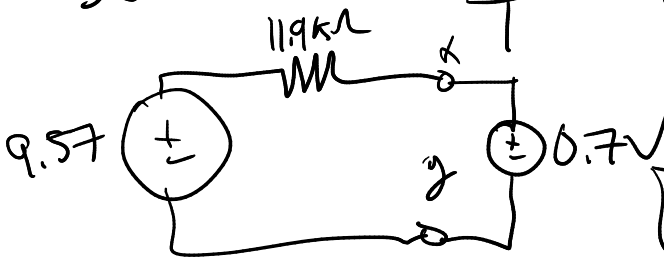
ask is $V < 0.7V$

$V = 9.57$, Diode must be ON

If we assume Diode is ON

Set $V = 0.7$ and  $\Rightarrow$ $0.7V$

Check if $i > 0$?




$= i \approx 745 \mu A$

$$\boxed{V = 0.7V, \; i = 745 \mu A}$$